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ON THE GENERALIZATION OF SFT: A REINFORCE-MENT LEARNING PERSPECTIVE WITH REWARD REC-

关于 SFT 的泛化：从强化学习视角看奖励重构

TIFICATION

通知

Yongliang Wu[1*] **Yizhou Zhou**[2*†] **Zhou Ziheng**[3] **Yingzhe Peng**[1] **Xinyu Ye**[4]
Xinting Hu[5] **Wenbo Zhu**[6] **Lu Qi**[7] **Ming-Hsuan Yang**[8] **Xu Yang**[1‡]
[1]Southeast University [2]Independent Researcher [3]University of California, Los Angeles
[4]Shanghai Jiao Tong University [5]Nanyang Technological University
[6]University of California, Berkeley
[7]Wuhan University [8]University of California, Merced yongliang0223@gmail.com, zyz0205@hotmail.com, xuyang_palm@seu.edu.cn

吴永亮 [1*] 王轶舟 [2*†] 周子恒 [3] 彭应哲 [1] 叶新宇 [4] 何新亭 [5] 朱文博 [6] 齐鲁 [7] 杨明轩 [8] 袁旭 [1‡]
[1] 东南大学 [2] 独立研究员 [3] 加州大学洛杉矶分校 [4] 上海交通大学 [5] 南洋理工大学 [6] 加州大学伯克利分校 [7] 武汉大学 [8] 加州默塞德分校 yongliang0223@gmail.com, zyz0205@hotmail.com, xuyang_palm@seu.edu.cn

ABSTRACT

摘要

In this work, we present a simple yet theoretically motivated improvement to Supervised Fine-Tuning (SFT) for the Large Language Model (LLM), address-ing its limited generalization compared to reinforcement learning (RL). Through mathematical analysis, we reveal that standard SFT gradients implicitly encode a problematic reward structure that may severely restrict the generalization ca-pabilities of model. To rectify this, we propose Dynamic Fine-Tuning (DFT), stabilizing gradient updates for each token by dynamically rescaling the objec-tive function with the probability of this token. With just a single-line change, the method outperforms standard SFT on multiple difficult benchmarks and base mod-els, from math reasoning to code generation and multi-modal tasks, demonstrating improved generalization. Additionally, DFT achieves competitive results in offline RL settings, providing an effective yet streamlined alternative. By bridging theo-retical insights with practical solutions, this work advances the state of SFT. The code will be available athttps://github.com/yongliang-wu/DFT.

在本工作中，我们提出了一种简单但具有理论依据的改进方法，用于大型语言模型（LLM）的监督微调（SFT），解决了其与强化学习（RL）相比泛化能力有限的问题。通过数学分析，我们揭示了标准 SFT 梯度隐式编码了一种有问题的奖励结构，这可能会严重限制模型的泛化能力。为了解决这个问题，我们提出了动态微调（DFT），通过动态地根据该标记的概率重新缩放目标函数，来稳定每个标记的梯度更新。只需一行代码的修改，该方法在多个困难基准测试和基

础模型上均优于标准 SFT，从数学推理到代码生成和多模态任务，展示了更好的泛化能力。此外，DFT 在离线强化学习设置中也取得了具有竞争力的结果，提供了一种有效且简洁的替代方案。通过将理论见解与实际解决方案相结合，本工作推动了 SFT 的发展。代码将在以下链接中提供：https://github.com/yongliang-wu/DFT.

1 INTRODUCTION

1 引言

Supervised Fine-Tuning (SFT), which adapts models to expert demonstrations, has become the stan-dard post-training paradigm for Large Language Models (LLMs). It enables efficient task adapta-tion and capability enhancement (Chung et al., 2024; Zhang et al., 2024b; Sanh et al., 2022; Ouyang et al., 2022), and is popular for its ease of implementation and rapid acquisition of expert-like behav-iors (Wei et al., 2022; Zhou et al., 2023). Despite these advantages, SFT often shows limited gener-alization compared to reinforcement learning (RL) (Chu et al., 2024; Ouyang et al., 2022; Christiano et al., 2017; Bai et al., 2022; Huan et al., 2025; Swamy et al., 2025). RL leverages explicit reward or verification signals to explore diverse strategies and thus generalizes better. However, RL requires substantial computation, careful hyperparameter tuning, and explicit reward signals—conditions of-ten impractical in real-world settings (Schulman et al., 2017; Ouyang et al., 2022; Sheng et al., 2025; Strubell et al., 2019; Liu & Yin, 2024; Winsta, 2025). Moreover, RL can struggle to recover expert-like behaviors that SFT captures efficiently (Mandlekar et al., 2022; Chen et al., 2025b).

监督微调（SFT）通过适应专家演示来调整模型，已成为大型语言模型（LLMs）的标准化后训练范式。它能够高效地进行任务适应和能力提升（Chung 等，2024；Zhang 等，2024b；Sanh 等，2022；Ouyang 等，2022），并且因其易于实现和快速获得类似专家行为而广受欢迎（Wei 等，2022；Zhou 等，2023）。尽管有这些优势，与强化学习（RL）相比，SFT 通常表现出有限的泛化能力（Chu 等，2024；Ouyang 等，2022；Christiano 等，2017；Bai 等，2022；Huan 等，2025；Swamy 等，2025）。RL 利用显式的奖励或验证信号来探索多样化的策略，因此泛化能力更强。然而，RL 需要大量的计算资源、仔细的超参数调优以及显式的奖励信号——这些条件在现实世界环境中往往难以实现（Schulman 等，2017；Ouyang 等，2022；Sheng 等，2025；Strubell 等，2019；Liu & Yin，2024；Winsta，2025）。此外，RL 在恢复 SFT 高效捕捉的类似专家行为方面可能会遇到困难（Mandlekar 等，2022；Chen 等，2025b）。

To exploit the complementary strengths of both approaches, many hybrid methods combine SFT with RL (Ouyang et al., 2022; Sheng et al., 2025; Rafailov et al., 2023; Liu et al., 2025; Qiu et al., 2025). Yet a key question remains: can SFT itself be fundamentally improved? This is crucial, as SFT remains the only viable option when datasets contain only positive demonstrations, with no negative samples or reward model available.

为了利用这两种方法的互补优势，许多混合方法将监督微调（SFT）与强化学习（RL）相结合（Ouyang 等，2022；Sheng 等，2025；Rafailov 等，2023；Liu 等，2025；Qiu 等，2025）。然而，一个关键问题仍然存在：SFT 本身是否能够从根本上得到改进？这一点至关重要，因为在数据集中仅包含正面示例而没有负面样本或奖励模型的情况下，SFT 仍然是唯一可行的选择。

In this work, we address this gap with a mathematical analysis of the connection between SFT and RL. We show that SFT's gradient update is equivalent to a policy gradient with a specific, implicitly defined reward. Crucially, this reward is (i) sparse, and (ii) inversely proportional to the model's

在本工作中，我们通过分析 SFT 与强化学习（RL）之间的联系的数学分析来解决这一问题。我们展示了 SFT 的梯度更新等价于一个具有特定、隐式定义奖励的策略梯度。关键的是，这种奖励具有以下两个特点：(i) 稀疏性，(ii) 与模型的某种属性成反比。

*Equal Contribution. †Project Leader. ‡Corresponding Author.

*同等贡献。†项目负责人。‡通讯作者。

probability of expert actions (see equation 6). As a result, when the model assigns low probability to expert actions, the gradient becomes excessively large—yielding an ill-posed reward structure and unstable, overfitting-prone optimization.

专家行为的概率（见公式 6）。因此，当模型对专家行为赋予较低概率时，梯度会变得过大——导致奖励结构不明确，并产生不稳定且容易过拟合的优化过程。

Building on this insight, we propose Dynamic Fine-Tuning (DFT), a principled fix. Our method rescales the SFT objective at each token by its probability, canceling the distortion introduced by inverse-probability weighting. This transforms the SFT gradient from an unstable, biased estimator into a stable, uniformly weighted update rule.

基于这一见解，我们提出了动态微调（DFT），这是一种有原则的解决方案。我们的方法在每个标记处根据其概率重新缩放 SFT 目标，从而抵消逆概率加权引入的失真。这将 SFT 梯度从一个不稳定的、有偏的估计量转变为一个稳定且均匀加权的更新规则。

Empirically, DFT delivers substantial improvements. On the Qwen-2.5-Math series (Qwen Team et al., 2024b) fine-tuned with NuminaMath-CoT (LI et al., 2024), DFT yields gains several times larger than standard SFT. More importantly, unlike SFT, which often degrades on challenging bench-marks such as OlympiadBench (He et al., 2024), AIME 2024 (American Institute of Mathematics, 2024), and AMC 2023 (Mathematical Association of America, 2023), our method consistently im-proves performance and generalization. These improvements hold across models, scales, and data sizes (Table 1, Figure 1), and extend to code generation and multimodal reasoning (Tables 3, 4).

从经验上看，DFT 带来了显著的改进。在使用 NuminaMath-CoT（LI 等，2024）微调的 Qwen-2.5-Math 系列（Qwen 团队等，2024b）模型上，DFT 的提升效果比标准的 SFT 高数倍。更重要的是，与经常在具有挑战性的基准测试（如 OlympiadBench（He 等，2024）、AIME 2024（美国数学学会，2024）和 AMC 2023（数学协会，2023））中表现不佳的 SFT 不同，我们的方法在性能和泛化能力上始终表现出提升。这些改进适用于各种模型、规模和数据量（见表 1 和图 1），并且也扩展到代码生成和多模态推理（见表 3 和表 4）。

We further test DFT in off-policy RL settings (Table 2), where dense rewards are available (Levine et al., 2020). Our method not only outperforms offline RL approaches such as DPO (Rafailov et al., 2023) and RAFT (Dong et al., 2023; Ahn et al., 2024), but also achieves competitive or superior performance to online methods like GRPO and PPO on math tasks with Qwen2.5-Math-1.5B. Unlike these RL methods, DFT requires neither a reference model nor large batch sizes, making it a simpler and more resource-efficient alternative.

我们进一步在离策略强化学习设置中测试 DFT（表 2），其中可以获取密集奖励（Levine 等，2020）。我们的方法不仅优于离线强化学习方法，如 DPO（Rafailov 等，2023）和 RAFT（Dong 等，2023；Ahn 等，2024），而且在使用 Qwen2.5-Math-1.5B 进行数学任务时，也能够达到与在线方法如 GRPO 和 PPO 相当甚至更优的性能。与这些强化学习方法不同，DFT 不需要参考模型或大批次大小，使其成为一种更简单且资源效率更高的替代方案。

To understand its effect, we analyze token probability distributions after training (Figure 2). While traditional SFT uniformly pushes probabilities toward the training set, DFT selectively increases some while reducing others. In particular, the proportion of less strongly fitted tokens rises, suggest-ing improved regularization and reduced overfitting. We provide further discussion in Appendix A.4.

为了理解其效果，我们分析训练后的标记概率分布（图 2）。虽然传统的 SFT 会将概率均匀地推向训练集，但 DFT 则有选择性地增加某些概率而减少其他概率。具体来说，拟合程度较低的标记所占比例上升，这表明正则化效果增强且过拟合现象减少。我们在附录 A.4 中进一步讨论了这一点。

The contributions of this work are theoretical and practical. On the theoretical side, we mathematically establish LLM SFT as a special RL in policy gradient space, pinpoint the underlying reasons for the limited generalization of SFT, and derive a method to improve it. On the experimental side, we show that such a simple solution, just one line of code, can enhance the performance and generalization capabilities of SFT across various tasks and models.

本工作的贡献既有理论上的，也有实践上的。在理论方面，我们数学上将大语言模型的监督微调（SFT）确立为策略梯度空间中的特殊强化学习（RL），指出了 SFT 泛化能力有限的潜在原因，并推导出一种改进方法。在实验方面，我们展示了这种简单的解决方案，只需一行代码，就能在各种任务和模型中提升 SFT 的性能和泛化能力。

2 Related Work

2 相关工作

The trade-off between supervised fine-tuning (SFT) and reinforcement learning (RL) is central to the alignment of large language models. SFT is widely adopted due to its simplicity and efficiency in imitating expert demonstrations (Chung et al., 2024; Zhou et al., 2023; Wei et al., 2022), analogous to behavioral cloning in robotics (Sammut, 2011; Mandlekar et al., 2022). However, the literature consistently highlights its limitations, particularly the tendency to overfit and generalize poorly compared to RL, which leverages reward signals to discover more robust policies (Ouyang et al., 2022; Christiano et al., 2017; Bai et al., 2022; Swamy et al., 2025; Zhang et al., 2025). A recent systematic comparison by Chu et al. (2024) across textual and visual domains confirms this distinction, concisely summarized as "SFT memorizes while RL generalizes." They further show that SFT remains indispensable as an initialization step, stabilizing output formatting prior to effective RL training. Nonetheless, RL faces significant practical hurdles, including computational expense, sensitivity to hyperparameters, and the requirement of an explicit reward function, all of which constrain its applicability (Schulman et al., 2017; Strubell et al., 2019; Sheng et al., 2025).

监督微调（SFT）与强化学习（RL）之间的权衡是大型语言模型对齐的核心。由于其在模仿专家示范方面的简单性和高效性，SFT 被广泛采用（Chung 等，2024；Zhou 等，2023；Wei 等，2022），类似于机器人领域的行为克隆（Sammut，2011；Mandlekar 等，2022）。然而，文献一致指出其局限性，特别是相较于利用奖励信号发现更鲁棒策略的 RL，SFT 容易过拟合且泛化能力较差（Ouyang 等，2022；Christiano 等，2017；Bai 等，2022；Swamy 等，2025；Zhang 等，2025）。Chu 等（2024）最近在文本和视觉领域进行的系统性比较进一步确认了这一区别，简明总结为“SFT 记忆而 RL 泛化”。他们进一步表明，SFT 仍然是初始化步骤中不可或缺的，它在有效 RL 训练之前稳定输出格式。尽管如此，RL 在实践中面临重大挑战，包括计算成本高、对超参数敏感以及需要显式奖励函数，这些因素都限制了其适用性（Schulman 等，2017；Strubell 等，2019；Sheng 等，2025）。

To combine the strengths of both paradigms, much recent work has pursued hybrid approaches. The most common strategy involves SFT pretraining followed by RL-based refinement with a learned reward model, as popularized by InstructGPT (Ouyang et al., 2022). More recent methods interleave SFT and RL updates to improve stability and performance (Sheng et al., 2025; Liu et al., 2025; Qiu et al., 2025). Other approaches, such as Direct Preference Optimization (DPO) (Rafailov et al., 2023), bypass reward modeling entirely by directly optimizing policies on preference data, thereby unifying imitation and reinforcement signals within a single loss function. Chen et al. (2025a) introduce Negative-aware Fine-Tuning (NFT), which models incorrect generations via an implicit negative policy, enabling self-improvement without explicit feedback. While powerful,

these meth-ods rely on reward signals, preference pairs, or negative samples. They enrich the training pipeline

为了结合两种范式的优点，许多近期的工作都致力于采用混合方法。最常见的策略是在 SFT（监督微调）预训练之后，使用基于强化学习的优化方法，并结合学习到的奖励模型，这种方法在 InstructGPT（Ouyang 等，2022）中得到了广泛推广。更近期的方法则交替进行 SFT 和强化学习的更新，以提高稳定性和性能（Sheng 等，2025；Liu 等，2025；Qiu 等，2025）。其他方法，例如直接偏好优化（DPO）（Rafailov 等，2023），则完全跳过奖励建模，而是直接在偏好数据上优化策略，从而在单一损失函数中统一模仿和强化信号。Chen 等（2025a）提出了负样本感知微调（NFT），通过隐式的负策略建模错误生成，从而在无需显式反馈的情况下实现自我提升。尽管这些方法功能强大，但它们依赖于奖励信号、偏好对或负样本。它们丰富了训练流程。

but do not fundamentally improve SFT in its native setting, where only positive demonstrations are available. Our work instead focuses on enhancing SFT itself without requiring external feedback.

但这些方法在原生设置下并未从根本上改进 SFT，因为在该设置中只能获得正面示例。相反，我们的工作专注于在不依赖外部反馈的情况下提升 SFT 本身。

A complementary line of theoretical research seeks to unify SFT and RL under a common formalism. Du et al. (2025) reinterpret RLHF as a reward-weighted variant of SFT, preserving reliance on an explicit reward. Wang et al. (2025) show that SFT can be cast as RL with an implicit reward, proposing adjustments such as smaller learning rates to manage the vanishing KL constraint. Abdol-maleki et al. (2025) analyze learning from both positive and negative feedback, studying how their balance affects convergence. Qin & Springenberg (2025) view SFT as a lower bound of RL and introduce importance weighting based on the data-generating policy. While these works establish connections between SFT and RL through weighting, they do not provide a precise mathematical equivalence between the SFT gradient and the offline policy gradient. Some methods approximate this connection in practice by reweighting training losses. For instance, MixCE (Zhang et al., 2023) combines the forward and reverse KL divergences to form a unified objective, while GOLD (Pang & He, 2021) adopts offline RL with demonstrations, introducing reliance on an unknown demonstra-tion distribution π_b and a restrictive $1/N$ assumption. In contrast, our work rigorously establishes this equivalence, pinpointing the inverse-probability weighting term in SFT as the key distinction. This observation directly motivates our proposed solution: multiplying the SFT loss by the model probability to eliminate the weighting.

一种补充性的理论研究路线旨在将 SFT 和 RL 统一在一个共同的框架下。Du 等（2025）重新诠释 RLHF 为一种带有奖励权重的 SFT 变体，保留了对显式奖励的依赖。Wang 等（2025）表明，SFT 可以视为具有隐式奖励的 RL，提出了如减小学习率等调整方法以管理消失的 KL 约束。Abdolmaleki 等（2025）分析了从正反馈和负反馈中学习的过程，研究了它们的平衡如何影响收敛性。Qin 和 Springenberg（2025）将 SFT 视为 RL 的一个下界，并基于数据生成策略引入了重要性加权。虽然这些工作通过加权方式建立了 SFT 和 RL 之间的联系，但并未提供 SFT 梯度与离线策略梯度之间精确的数学等价性。一些方法通过重新加权训练损失来近似这种联系。例如，MixCE（Zhang 等，2023）结合正向和反向 KL 散度形成统一的目标函数，而 GOLD（Pang 和 He，2021）采用带有演示的离线 RL，引入了对未知演示分布 π_b 和限制性的 1/N 假设的依赖。相比之下，我们的工作严格建立了这种等价性，指出了 SFT 中的逆概率加权项是关键区别。这一观察直接启发了我们提出的解决方案：通过将 SFT 损失乘以模型概率来消除加权。

Interestingly, our method modifies the standard cross-entropy (CE) loss in a way that inverts the weighting philosophy of the widely used Focal Loss (Lin et al., 2017). Specifically, our modified CE takes the form $-p\log(p)$, whereas focal loss is defined as $-(1-p)^{\gamma}\log(p)$. Focal Loss deliberately downweights well-classified samples to emphasize underrepresented or hard cases, whereas we deliberately downweight poorly classified samples to encourage generalization. This inversion reflects a fundamental shift in the LLM era: while underfitting was once a central challenge, overfitting and memorization now dominate, demanding a rethinking of objective design.

有趣的是，我们的方法以一种方式修改了标准的交叉熵（CE）损失，从而反转了广泛使用的 Focal Loss（Lin 等，2017）所采用的加权哲学。具体来说，我们的修改后的 CE 形式为$-p\log(p)$，而 Focal Loss 定义为$-(1-p)^{\gamma}\log(p)$。Focal Loss 有意降低分类效果良好的样本的权重，以强调那些代表性不足或困难的样本，而我们则有意降低分类效果差的样本的权重，以鼓励模型的泛化能力。这种反转反映了 LLM 时代的基本转变：虽然过去欠拟合是主要挑战，但现在过拟合和记忆效应占主导地位，这要求我们重新思考目标函数的设计。

3 METHOD

3 方法

3.1 PRELIMINARIES

3.1 基础知识

Supervised Fine-Tuning. Let $D = \{(x, y_\star)\}$ denote a corpus of expert demonstrations, where $y_\star$ is the complete reference response to the query x. SFT minimizes the sentence-level cross-entropy:

监督微调。设 $D = \{(x, y_\star)\}$ 表示专家演示的语料库，其中 $y_\star$ 是查询 x 的完整参考回答。SFT 最小化句子级别的交叉熵：

$$L_{\mathrm{SFT}}(\theta) = \mathrm{E}_{(x,y_\star)\sim D} - \log \pi_\theta y_\star \mid x \,. \qquad (1)$$

$$L_{SFT}(\theta) = E_{(x,y_\star)\sim D}[-\log \pi_\theta(y_\star|x)] \quad (1)$$

Its gradient is:

$$\nabla_\theta L_{\mathrm{SFT}}(\theta) = \mathrm{E}_{(x,y_\star)\sim D} - \nabla_\theta \log \pi_\theta y_\star \mid x \,. \qquad (2)$$

其梯度为：

$$\nabla_\theta L_{SFT}(\theta) = E_{(x,y_\star)\sim D}[-\nabla_\theta \log \pi_\theta(y_\star|x)] \,. \quad (2)$$

Reinforcement Learning. Let y denote a response sampled from the policy $\pi_\theta(\cdot \mid x)$ for query x. Given a reward function $r(x, y) \in \mathrm{R}$, the policy objective is

强化学习。令 y 表示从策略 $\pi_\theta(\cdot \mid x)$ 对查询 x 采样的响应。给定一个奖励函数 $r(x, y) \in \mathrm{R}$，策略的目标是

$$J(\theta) = \mathrm{E}_{x\sim D_x,\, y\sim\pi_\theta(\cdot|x)}\, r(x, y) \,. \quad (3)$$

$$J(\theta) = E_{x\sim D_x,\, y\sim \pi_\theta(\cdot|x)}[r(x, y)] \,. \quad (3)$$

Its policy gradient at the sentence level is

$$\nabla_\theta J(\theta) = \mathrm{E}_{x\sim D_x,\, y\sim\pi_\theta(\cdot|x)}\, \nabla_\theta \log \pi_\theta(y \mid x)\, r(x, y) \,. \qquad (4)$$

其句子级别的策略梯度为

$$\nabla_\theta J(\theta) = E_{x\sim D_x,\, y\sim \pi_\theta(\cdot|x)} \nabla_\theta \log \pi_\theta(y \mid x)\, r(x, y) \,. \qquad (4)$$

3.2 UNIFY SFT AND RL GRADIENT EXPRESSION

3.2 统一 SFT 和 RL 梯度表达式

Rewriting SFT Gradient as Policy Gradient via Importance Sampling. The SFT gradient in equation 2 is taken under the *fixed* demonstration distribution. We convert it to an on-policy expectation by inserting an importance weight that compares the expert (Dirac Delta) distribution with the model distribution.

通过重要性采样将 SFT 梯度重写为策略梯度。公式 2 中的 SFT 梯度是在固定演示分布下计算的。我们通过插入一个重要性权重，将该梯度转换为基于策略的期望值，该权重比较了专家（狄拉克 δ）分布与模型分布。

$$\mathrm{E}_{(x,y_\star)\sim D} - \nabla_\theta \log \pi_\theta y_\star \mid x = \mathrm{E}_{x\sim D_x}\, \mathrm{E}_{y\sim\pi_\theta(\cdot|x)} \underbrace{\frac{\mathbf{1}[y = y_\star]}{\pi_\theta(y \mid x)}}_{\{z} - \nabla_\theta \log \pi_\theta y \mid x \qquad (5)$$

Define the auxiliary variables (importance sampling weight) as

$$w(y \mid x) = \frac{1}{\pi_\theta(y \mid x)}, \qquad r(x, y) = \mathbf{1}[y = y^\star].$$

定义辅助变量（重要性采样权重）为

$$
w(y \mid x) = \frac{1}{\pi_\theta(y \mid x)}, \quad r(x, y) = \mathbf{1}[y = y^\star].
$$

Reorganizing equation 5 and rewriting it using the above auxiliary variables, we obtain the form

重新整理方程 5，并使用上述辅助变量对其进行重写，我们得到如下形式

$$\nabla_\theta L_{\mathrm{SFT}}(\theta) = -\mathbb{E}_{x\sim D_x,\, y\sim\pi_\theta(\cdot|x)}\, w(y \mid x)\ \nabla_\theta \log \pi_\theta(y \mid x)\ r(x, y)\ . \quad (6)$$

This form of the SFT gradient now aligns closely with the policy gradient equation 4. Thus we can see, *conventional SFT is precisely an on-policy-gradient with the reward as an indicator func-tion of matching the expert trajectory but biased by an importance weighting* $1/\pi_\theta$. More detailed derivations can be found in Appendix A.3.

这种 SFT 梯度形式现在与策略梯度方程 4 非常接近。因此我们可以看到，传统的 SFT 实际上是一种基于策略的梯度，其中奖励作为匹配专家轨迹的指示函数，但受到重要性加权 1/π θ 的偏差影响。更详细的推导请参见附录 A.3。

Given the unavoidable sparsity of reward signals in the SFT setting, we identify the importance sampling weight $1/\pi_\theta$ as a fundamental cause of the poor generalization of SFT relative to RL. When the model assigns low probability to the expert response, the weight w becomes excessively large, leading to an ill-posed reward structure. In such cases, the gradient can grow disproportion-ately, making the training unstable and prone to overfitting. This issue is further exacerbated by the extreme sparsity of the reward function, since $r(x, y) = \mathbf{1}[y = y_\star]$ is non-zero only when the model exactly matches the expert output. As a result, optimization tends to overfit rare exact-match demonstrations, undermining the model's ability to generalize beyond training data.

鉴于在 SFT 设置中奖励信号不可避免的稀疏性，我们识别出重要性采样权重 1/π θ 是 SFT 相对于强化学习（RL）泛化能力差的根本原因。当模型对专家响应分配的低概率时，权重 w 会变得过大，导致奖励结构不明确。在这种情况下，梯度可能会不成比例地增长，使训练变得不稳定并容易过拟合。这一问题进一步被奖励函数的极端稀疏性所加剧，因为 r(x, y) = 1[y = y⋆]只有在模型恰好与专家输出匹配时才非零。因此，优化往往会过度拟合罕见的精确匹配示例，从而削弱模型在训练数据之外泛化的能力。

3.3 Proposed Method

3.3 提出的方法

Reward Rectification via Dynamic Reweighting. To neutralize the skewed reward issue identi-fied when viewing SFT under the RL objective, we dynamically reweight the reward by multiplying by a corrective inverse ratio given by the policy probability $1/w$. The resulting "dynamically fine-tuned" gradient is then

通过动态重加权进行奖励修正。为了中和在以强化学习目标看待 SFT 时发现的奖励偏差问题，我们通过乘以由策略概率 1/w 给出的校正逆比值来动态地重新加权奖励。由此得到的“动态微调”梯度随后是

$$\nabla_\theta L_{\mathrm{DFT}}(\theta) = \nabla_\theta L_{\mathrm{SFT}}(\theta) \cdot \mathrm{sg}(\ 1\ w) = \nabla_\theta L_{\mathrm{SFT}}(\theta) \cdot \mathrm{sg}(\pi_\theta(y_\star \mid x)). \quad (7)$$

where sg(·) denotes the stop gradient operator, ensuring that gradients do not flow through the reward scaling term w. To facilitate transition to later equations, we directly write $1/w$ to be $\pi_\theta(y_\star \mid x)$

instead of $\pi_\theta(y \mid x)$ because the indicator function in equation 5 or equation 6 would leave all cases where $y \neq y_\star$ is 0. Now since the gradient does not flow, the corrected SFT loss also becomes a simple reweighted loss, called Dynamic Fine-tuning (DFT).

其中，sg(・) 表示停止梯度操作，确保梯度不会通过奖励缩放项 w 流动。为了便于后续方程的转换，我们直接将 1/w 写作 $\pi_\theta(y_\star|x)$，而不是 $\pi_\theta(y|x)$，因为方程 5 或方程 6 中的指示函数会使所有 $y \neq y_\star$ 的情况为 0。现在由于梯度不再流动，修正后的 SFT 损失也变成了一个简单的加权损失，称为动态微调（DFT）。

$$L_{\mathrm{DFT}}(\theta) = \mathrm{E}_{(x,y_\star)\sim D}\left[\mathrm{sg}\,\pi_\theta(y_\star|x)\log \pi_\theta(y_\star|x)\right]. \tag{8}$$

However, in practice, computing importance weights over the entire trajectory can induce numerical instability. A common treatment of this issue is to simply apply importance sampling at the token level, as was adopted in PPO (Schulman et al., 2017). This leads to the final DFT loss version:

然而，在实践中，对整个轨迹计算重要性权重可能会导致数值不稳定。解决这一问题的常见方法是在 token 级别应用重要性采样，如 PPO（Schulman 等，2017）所采用的那样。这导致了最终的 DFT 损失版本：

$$L_{\mathrm{DFT}}(\theta) = \mathrm{E}_{(x,y_\star)\sim D}\left[\sum_{t=1}^{|y_\star|}\mathrm{sg}\,\pi_\theta(y_{\star t}|y_{\star<t},x)\log \pi_\theta(y_{\star t}|y_{\star<t},x)\right]. \tag{9}$$

LDFT(θ) = E_{(x,y⋆)~D} [sg π θ (y⋆t| y⋆<t, x) log π θ (y⋆t| y⋆<t, x)] . (9)

Note that the reward of this corrected SFT (in RL form), i.e., DFT, now becomes 1 uni-formly for all expert trajectory. This is akin to contemporary verification based reward approach RLVR (DeepSeek-AI et al., 2025) that assigns uniform reward to all correct samples. Consequently, it avoids over-concentration on specific low-probability reference tokens, leading to more stable updates and improved generalization without introducing any additional sampling or reward models.

请注意，这种修正后的 SFT（以强化学习形式）的奖励，即 DFT，现在对所有专家轨迹都统一为 1。这类似于当前基于验证的奖励方法 RLVR（DeepSeek-AI 等，2025）的做法，即对所有正确样本赋予相同的奖励。因此，这种方法避免了对特定低概率参考标记的过度集中，从而实现了更稳定的更新和更好的泛化能力，而无需引入任何额外的采样或奖励模型。

4 Experiments

4 实验

We design four groups of experiments to comprehensively evaluate DFT. We first study the standard SFT setting on mathematical reasoning tasks to establish its core advantage over SFT (Section 4.1). We then extend to an offline RL setting, comparing DFT with representative offline and online RL methods (Section 4.2). To test cross-domain robustness, we further examine DFT on code generation benchmarks (Section 4.3) and its applicability to multi-modal reasoning math datasets (Section 4.4).

我们设计了四组实验，全面评估 DFT。我们首先在数学推理任务上研究标准的 SFT 设置，以确立其相对于 SFT 的核心优势（第 4.1 节）。然后我们扩展到离线强化学习（offline RL）设

置，将 DFT 与具有代表性的离线和在线强化学习方法进行比较（第 4.2 节）。为了测试跨领域鲁棒性，我们进一步在代码生成基准测试上检验 DFT（第 4.3 节），并评估其在多模态推理数学数据集上的适用性（第 4.4 节）。

Table 1: Average@16 accuracy of five state-of-the-art large language models on mathematical reasoning benchmarks. The best performance of each model across benchmarks is bold.

表 1：五种最先进的大型语言模型在数学推理基准测试中的平均@16 准确率。每个模型在各个基准测试中的最佳表现用粗体表示。

	Math500	Minerva Math	Olympiad Bench	AIME24	AMC23	Avg.
LLaMA-3.2-3B	1.63	1.36	1.01	0.41	1.56	1.19
LLaMA-3.2-3B w/SFT	8.65	2.38	2.06	0.00	3.13	3.24
LLaMA-3.2-3B w/DFT	**12.79**	**2.84**	**2.90**	**0.83**	**3.91**	**4.65**
LLaMA-3.1-8B	1.86	0.98	0.94	0.21	1.01	1.00
LLaMA-3.1-8B w/SFT	16.85	5.78	3.88	0.00	5.16	6.33
LLaMA-3.1-8B w/DFT	**27.44**	**8.26**	**6.94**	**0.41**	**12.03**	**11.02**
DeepSeekMath-7B	6.15	2.15	1.74	0.21	2.97	2.64
DeepSeekMath-7B w/SFT	26.83	7.26	6.33	0.41	8.28	9.82
DeepSeekMath-7B w/DFT	**41.46**	**16.79**	**15.00**	**1.24**	**16.25**	**18.15**
Qwen2.5-Math-1.5B	31.66	8.51	15.88	4.16	19.38	15.92
Qwen2.5-Math-1.5B w/SFT	43.76	13.04	12.63	1.87	18.75	18.01
Qwen2.5-Math-1.5B w/DFT	**64.89**	**20.94**	**27.08**	**6.87**	**38.13**	**31.58**
Qwen2.5-Math-7B	40.12	14.39	17.12	6.68	27.96	21.25
Qwen2.5-Math-7B w/SFT	53.96	16.66	18.93	2.48	26.09	23.62
Qwen2.5-Math-7B w/DFT	**68.20**	**30.16**	**33.83**	**8.56**	**45.00**	**37.15**

4.1 Main Experiment - Mathematical Reasoning Task

To examine whether DFT can outperform vanilla SFT across tasks, architectures, and scales, we use mathematical reasoning as a representative testbed.

为了检验 DFT 是否在各种任务、架构和规模上都能优于传统的 SFT，我们使用数学推理作为代表性的测试平台。

4.1.1 Setup and Implementation Details

4.1.1 设置与实施细节

To efficiently manage computational resources, We andomly sample 100,000 instances from the the NuminaMath-CoT dataset (LI et al., 2024) for training. We conduct experiments using multiple models, including Qwen2.5-Math-1.5B, Qwen2.5-Math-7B (Qwen Team et al., 2024a), LLaMA-3.2-3B, LLaMA-3.1-8B (Dubey et al., 2024), and DeepSeekMath-7B (Shao et al., 2024).

为了高效管理计算资源，我们从 NuminaMath-CoT 数据集（LI 等，2024）中随机选取 100,000 个实例进行训练。我们使用多种模型进行实验，包括 Qwen2.5-Math-1.5B、Qwen2.5-Math-7B（Qwen 团队，2024a）、LLaMA-3.2-3B、LLaMA-3.1-8B（Dubey 等，2024）以及 DeepSeekMath-7B（Shao 等，2024）。

Our implementation builds upon the verl framework (Sheng et al., 2025), using recommended SFT hyper-parameters. Specifically, we employ the AdamW optimizer with learning rates of $5\times10^{-5}$for all models except the LLaMA-3.1-8B-Base, for which we adopt a lower learning rate of 2×10^{-5}. We set the mini-batch size to 256 and the maximum input length to 2048 tokens. The learning rate follows a cosine decay schedule with a warm-up ratio of 0.1. We evaluate on benchmarks including Math500 (Hendrycks et al., 2021), Minerva Math (Lewkowycz et al., 2022), Olympiad Bench (He et al., 2024), AIME 2024 (American Institute of Mathematics, 2024), and AMC 2023 (Mathematical Association of America, 2023) through the official Qwen2.5-Math evaluation pipeline (Qwen Team et al., 2024a). Each model uses the default chat template and Chain-of-Thought (CoT) prompting to stimulate step-by-step reasoning. All reported results represent average accuracy across 16 decoding runs, evaluated with a temperature of 1.0 and maximum generation length of 4096 tokens.

我们的实现基于 verl 框架（Sheng 等，2025），使用推荐的 SFT 超参数。具体来说，我们采用 AdamW 优化器，所有模型的学习率均为 5×10^{-5} ，除了 LLaMA-3.1-8B-Base，我们采用较低的学习率 2×10^{-5} 。我们将小批量大小设置为 256，最大输入长度为 2048 个标记。学习率遵循余弦衰减计划，并带有 0.1 的预热比例。我们通过官方的 Qwen2.5-Math 评估流程（Qwen 团队等，2024a）在多个基准测试上进行评估，包括 Math500（Hendrycks 等，2021）、Minerva Math（Lewkowycz 等，2022）、Olympiad Bench（He 等，2024）、AIME 2024（美国数学学会，2024）和 AMC 2023（美国数学协会，2023）。每个模型均使用默认的聊天模板和思维链（CoT）提示来激发逐步推理。所有报告的结果均表示在 16 次解码运行中的平均准确率，评估时温度为 1.0，最大生成长度为 4096 个标记。

4.1.2 Main Results

4.1.2 主要结果

DFT consistently yields average performance improvements over base models compared to standard SFT across all benchmarks. Table 1 shows that, for Qwen2.5-Math-1.5B, DFT achieves an average gain of +15.66 points over the base model, which is over 5.9× larger than the +2.09 point improve-ment from SFT. This pattern generalizes across other model families and sizes: LLaMA-3.2-3B benefits from a +3.46 point gain with DFT, exceeding the SFT gain (+2.05) by approximately 1.4×; LLaMA-3.1-8B achieves +10.02 from DFT, surpassing SFT's +5.33 by 1.88×; DeepSeekMath-7B sees a +15.51 point improvement via DFT, which is 1.58× larger than SFT's +7.18; and Qwen2.5-Math-7B reaches a +15.90 point gain, nearly 3.8× higher than the SFT improvement of +2.37.

DFT 在所有基准测试中相比标准 SFT 始终表现出比基线模型更高的平均性能提升。表 1 显示，对于 Qwen2.5-Math-1.5B，DFT 在基线模型上实现了平均提升+15.66 分，这一提升比 SFT 的+2.09 分提升高出 5.9 倍以上。这种模式适用于其他模型家族和规模：LLaMA-3.2-3B 通过 DFT 获得+3.46 分的提升，超过 SFT 的+2.05 分提升约 1.4 倍；LLaMA-3.1-8B 通过 DFT 获得+10.02 分的提升，超过 SFT 的+5.33 分提升 1.88 倍；DeepSeekMath-7B 通过 DFT 获得+15.51 分的提升，比 SFT 的+7.18 分提升高出 1.58 倍；Qwen2.5-Math-7B 达到+15.90 分的提升，几乎比 SFT 的+2.37 分提升高出 3.8 倍。

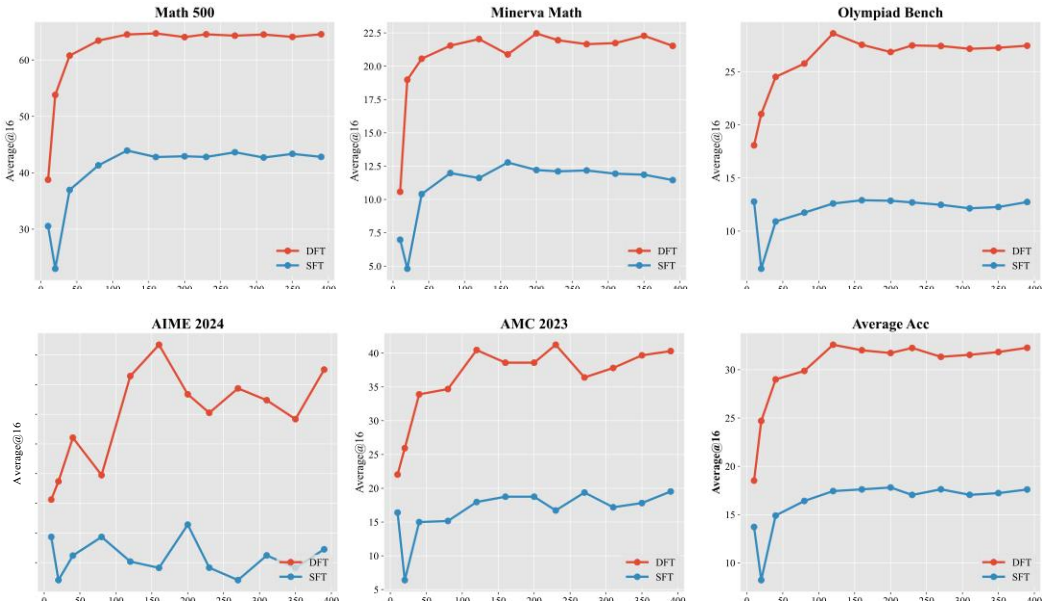

Figure 1: Accuracy progression for Qwen2.5-Math-1.5B across mathematical benchmarks, illustrating faster convergence and better performance achieved by DFT relative to SFT.

图 1：Qwen2.5-Math-1.5B 在数学基准测试中的准确性提升情况，展示了 DFT 相对于 SFT 实现更快的收敛速度和更好的性能。

DFT demonstrates generalization and robustness, especially on challenging benchmarks where stan-dard SFT yields minimal or even negative impact. For instance, on Olympiad Bench, SFT degrades performance for Qwen2.5-Math-1.5B, dropping accuracy from 15.88 to 12.63, while DFT boosts it to 27.08, +11.20 point improvement over base model. On AIME24, SFT reduces accuracy for Qwen2.5-Math-7B by 4.20 points (from 6.68 to 2.48), whereas DFT improves performance to 8.56, achieving a +1.88 point gain over the base model despite the difficulty of the benchmark. A similar trend is observed on AMC23. SFT reduces the performance of Qwen2.5-Math-1.5B from 19.38 to 18.75, while DFT raises it to 38.13, a +18.75 point gain over base. For Qwen2.5-Math-7B, SFT yields only a marginal improvement (+1.86), whereas DFT achieves a +17.04 point gain. These results underscore that DFT not only scales more effectively across models of varying capacities, but also exhibits better resilience on difficult reasoning tasks where traditional SFT struggles.

DFT 展示了良好的泛化能力和鲁棒性，尤其是在标准 SFT 在挑战性基准测试上效果有限甚至产生负面影响的情况下。例如，在奥林匹亚基准测试（Olympiad Bench）中，SFT 使 Qwen2.5-Math-1.5B 的性能下降，准确率从 15.88 降至 12.63，而 DFT 则将其提升至 27.08，比基线模型提升了 11.20 个百分点。在 AIME24 测试中，SFT 使 Qwen2.5-Math-7B 的准确率下降了 4.20 个百分点（从 6.68 降至 2.48），而 DFT 则将性能提升至 8.56，尽管基准测试难度较高，但仍比基线模型提升了 1.88 个百分点。在 AMC23 测试中也观察到了类似的趋势。SFT 使 Qwen2.5-Math-1.5B 的性能从 19.38 降至 18.75，而 DFT 则将其提升至 38.13，比基线模型提升了 18.75 个百分点。对于 Qwen2.5-Math-7B，SFT 仅带来小幅提升（+1.86），而 DFT

则提升了 17.04 个百分点。这些结果表明，DFT 不仅在不同容量的模型上具有更有效的扩展能力，而且在传统 SFT 难以应对的复杂推理任务中表现出更强的鲁棒性。

DFT exhibits better learning efficiency and faster convergence characteristics. Figure 1 reveals clear differences in learning dynamics between DFT and standard SFT on Qwen2.5-Math-1.5B across all math reasoning benchmarks. Compared to SFT, our method demonstrates three distinct advantages: (1) Faster convergence, achieving peak performance within the first 120 training steps on most benchmarks; (2) Better early-stage performance, with DFT already outperforming best final accuracy of SFT within the first 10–20 steps; and (3) Higher sample efficiency, consistently requiring fewer updates to reach relatively optimal results. This accelerated convergence shows that the dynamic reweighting mechanism in DFT leads to more informative gradient updates, guiding the model toward high-quality solutions early in training. It also suggests that DFT helps avoid the optimization plateaus or noise-prone regions often encountered in standard SFT, thereby enabling more efficient acquisition of complex mathematical reasoning patterns.

DFT 表现出更好的学习效率和更快的收敛特性。图 1 展示了 DFT 与标准 SFT 在 Qwen2.5-Math-1.5B 上在所有数学推理基准测试中的学习动态差异。与 SFT 相比，我们的方法具有三个显著优势：（1）更快的收敛速度，在大多数基准测试中，只需前 120 个训练步骤即可达到峰值性能；（2）早期阶段表现更好，DFT 在前 10-20 个步骤内就已经超过了 SFT 的最佳最终准确率；（3）更高的样本效率，能够以更少的更新次数达到相对最优的结果。这种加速收敛表明，DFT 中的动态重加权机制带来了更具信息量的梯度更新，引导模型在训练初期向高质量解靠拢。这也表明，DFT 有助于避免标准 SFT 中常遇到的优化平台期或噪声较大的区域，从而更高效地获取复杂的数学推理模式。

We also report the results of parameter-efficient fine-tuning (PEFT) training setting (Hu et al., 2022) and training on the OpenR1-Math dataset (Hugging Face, 2025) with better quality in Ap-pendix A.7 and Appendix A.6, respectively. Comparison and Discussion with the concurrent method iw-SFT (Qin & Springenberg, 2025) is provided in Appendix A.5.

我们在附录 A.7 和附录 A.6 中分别报告了参数高效微调（PEFT）训练设置（Hu 等，2022）以及在 OpenR1-Math 数据集（Hugging Face，2025）上的训练结果，该数据集具有更高的质量。与同期方法 iw-SFT（Qin & Springenberg，2025）的比较和讨论见附录 A.5。

4.2 Exploratory Experiment - Offline RL Setting

4.2 探索性实验 – 离线强化学习设置

Equation 7 shows that SFT suffers from reward sparsity, since in a constructed dataset each query x has only a single reference answer $y_\star$. From the perspective of RL, RFT/RAFT (Dong et al., 2023; Ahn et al., 2024) can be viewed as alleviating the sparse reward issue by effectively increasing

第 7 式表明，SFT 存在奖励稀疏的问题，因为在构造的数据集中，每个查询 x 只有一个参考答案 $y_\star$。从强化学习的角度来看，RFT/RAFT（Dong 等，2023；Ahn 等，2024）可以通过有效增加奖励密度来缓解这一问题。

Table 2: Evaluation results on mathematical reasoning benchmarks in an offline reinforcement learn-ing setting using reward signals from rejection sampling. The best performance is in bold.

表 2：在离线强化学习设置下，使用拒绝抽样奖励信号对数学推理基准进行评估的结果。最佳性能以粗体显示。

	Setting	Math500	Minerva Math	Olympiad Bench	AIME24	AMC23	Avg.
Qwen2.5-Math-1.5B	–	31.66	8.51	15.88	4.16	19.38	15.92
Qwen2.5-Math-1.5B w/DPO	Offline	46.89	11.53	22.86	4.58	30.16	23.20
Qwen2.5-Math-1.5B w/RFT	Offline	48.23	14.19	22.29	4.37	30.78	23.97
Qwen2.5-Math-1.5B w/PPO	Online	56.10	15.41	26.33	7.50	37.97	28.66
Qwen2.5-Math-1.5B w/GRPO	Online	62.86	18.93	28.62	8.34	41.25	32.00
Qwen2.5-Math-1.5B w/DFT	Offline	**64.71**	**25.16**	**30.93**	**7.93**	**48.44**	**35.43**

reward density, thereby enhancing model performance. Motivated by this observation, we conduct an exploratory study applying DFT in an offline RL setting, where the reward sparsity problem is inherently less severe compared to standard SFT, to further validate the effectiveness.

奖励密度，从而提高模型性能。受这一观察的启发，我们在离线强化学习（offline RL）环境中进行探索性研究，应用 DFT，因为在标准 SFT 中，奖励稀疏性问题比标准 SFT 更不严重，以进一步验证其有效性。

4.2.1 Setup and Implementation Details

4.2.1 设置与实现细节

We sample responses for 100,000 math questions using a temperature of 1.0 and generate four responses per question from the base model itself. Correct responses are identified using math verify and retained as training data, resulting in approximately 140,000 examples. For DPO training, we construct 100,000 positive–negative preference pairs from the generated responses.

我们使用温度为 1.0 对 100,000 个数学问题进行抽样，并从基础模型自身生成每个问题的四个回答。通过 math verify 识别正确回答，并将其保留作为训练数据，最终得到大约 140,000 个示例。对于 DPO 训练，我们从生成的回答中构建 100,000 个正负偏好对。

We compare DFT with representative offline RL methods, including DPO (Rafailov et al., 2023) and RFT (Dong et al., 2023; Ahn et al., 2024), as well as online RL methods PPO (Schulman et al., 2017) and GRPO (Shao et al., 2024). For RFT and DFT, the training setup follows the configuration in Section 4.1. For DPO, we use the ms-swift (Zhao et al., 2024) with a learning rate of 1×10^{-6}, batch size of 128, and a warmup ratio of 0.05. For PPO and GRPO, training is performed using the verl (Sheng et al., 2025) with a learning rate of 1×10^{-6}, batch size of 256, and a warmup ratio of 0.1. We set the number of response $n = 4$ for GRPO.

我们比较了 DFT 与具有代表性的离线强化学习方法，包括 DPO（Rafailov 等，2023）和 RFT（Dong 等，2023；Ahn 等，2024），以及在线强化学习方法 PPO（Schulman 等，2017）和 GRPO（Shao 等，2024）。对于 RFT 和 DFT，训练设置遵循第 4.1 节中的配置。对于 DPO，我们使用 ms-swift（Zhao 等，2024），学习率为 1×10^{-6}，批量大小为 128，预热比为 0.05。对于 PPO 和 GRPO，训练使用 verl（Sheng 等，2025），学习率为 1×10^{-6}，批量大小为 256，预热比为 0.1。我们为 GRPO 设置响应数量 $n = 4$。

4.2.2 Results

4.2.2 结果

DFT demonstrates competitive performance in the offline RL setting, outperforming both offline and online RL baselines. Table 2 shows DFT achieves an average score of 35.43, exceeding the best offline method RFT by +11.46 points, and even outperforming the strongest online RL algorithm GRPO by +3.43 points. Specially, on Math500, DFT scores 64.71, slightly ahead of GRPO (62.86) and better than PPO (56.10) and RFT (48.23). The gains are also notable on more challenging benchmarks: on AMC23, DFT achieves 48.44, a +7.19 point margin over GRPO and a +17.66 point gain over RFT. Similarly, on Minerva Math, DFT reaches 25.16, outperforming GRPO by +6.23 points, PPO by +9.75, and all offline baseline methods.

DFT 在离线强化学习设置中表现出色，优于离线和在线强化学习基线。表 2 显示，DFT 实现了平均分数 35.43，超过了最佳离线方法 RFT 约+11.46 分，甚至超过了最强的在线强化学习算法 GRPO 约+3.43 分。特别地，在 Math500 上，DFT 得分 64.71，略微领先于 GRPO（62.86），并且优于 PPO（56.10）和 RFT（48.23）。在更具挑战性的基准测试中，DFT 的提升也十分显著：在 AMC23 上，DFT 获得 48.44 分，比 GRPO 高出+7.19 分，比 RFT 高出+17.66 分。同样地，在 Minerva Math 上，DFT 达到 25.16 分，比 GRPO 高出+6.23 分，比 PPO 高出+9.75 分，并且优于所有离线基线方法。

These results highlight the strength of DFT as a simple yet effective fine-tuning strategy. Despite its lack of iterative reward modeling or environment interaction, it provides a stronger learning signal than both offline methods like DPO/RFT and online policy optimization algorithms like PPO/GRPO in certain scale train set. This suggests that DFT can serve as a more efficient and scalable alternative to traditional RL pipelines, particularly in domains where preference supervision is available but reward modeling or online response sampling is expensive or impractical.

这些结果突显了 DFT 作为简单而有效的微调策略的优势。尽管它缺乏迭代奖励建模或环境交互，但在某些规模的训练集上，它提供的学习信号比离线方法（如 DPO/RFT）和在线策略优化算法（如 PPO/GRPO）更强。这表明 DFT 可以作为传统强化学习流程更高效且可扩展的替代方案，特别是在有偏好监督但奖励建模或在线响应采样成本高昂或不切实际的领域中。

4.3 Exploratory Experiment - Code Generation Task

4.3 探索性实验 – 代码生成任务

To further validate the effectiveness of DFT, we investigate its performance on another reasoning-intensive task: code generation.

为进一步验证 DFT 的有效性，我们研究了它在另一个需要大量推理的任务——代码生成中的表现。

4.3.1 Setup and Implementation Details

4.3.1 设置与实现细节

We adopt UltraFeedback (Cui et al., 2024) as the training dataset. From this corpus, we sample 10,000 prompts and, for each prompt, select the response with the highest average score to perform supervised fine-tuning (SFT) (Du et al., 2025). Model performance is assessed on three widely used code generation benchmarks: HumanEval (Chen et al., 2021), HumanEval+ (Liu et al., 2023),

我们采用 UltraFeedback（Cui 等，2024）作为训练数据集。从该语料库中，我们采样了 10,000 个提示，并为每个提示选择平均得分最高的响应来进行监督微调（SFT）（Du 等，2025）。模

型性能在三个广泛使用的代码生成基准上进行评估：HumanEval（Chen 等，2021）、HumanEval+（Liu 等，2023）

Table 3: Performance of various models on code generation benchmarks. The best performance for each benchmark is highlighted in bold.

表 3：各种模型在代码生成基准测试中的性能。每个基准测试中表现最好的结果加粗显示。

	HumanEval		MultiPL-E								
	HumanEval	**HumanEval+**	**Python**	**C++**	**Java**	**PHP**	**TS**	**C#**	**Bash**	**JS**	**Avg.**
Qwen2.5-3B	43.3	36.0	43.29	40.99	37.34	37.89	**47.17**	**43.04**	24.68	45.96	40.05
Qwen2.5-3B w/SFT	41.5	34.8	42.07	42.24	37.97	37.27	43.40	41.77	20.25	**47.83**	39.10
Qwen2.5-3B w/DFT	**45.7**	**39.0**	**45.73**	**44.72**	**41.77**	**45.34**	42.14	**43.04**	**27.85**	44.10	**41.84**
Qwen2.5-Coder-3B	52.4	42.7	51.83	53.42	46.20	47.20	54.09	55.06	25.32	**54.04**	48.39
Qwen2.5-Coder-3B w/SFT	51.8	43.9	51.22	51.55	48.10	54.66	**59.12**	51.27	**34.18**	**54.04**	50.52
Qwen2.5-Coder-3B w/DFT	**56.7**	**50.0**	**57.32**	**54.66**	**51.27**	**58.39**	58.49	**60.76**	31.01	53.42	**53.16**
Qwen2.5-Coder-7B	62.2	53.0	63.41	63.98	53.16	59.01	62.89	59.49	39.24	60.87	57.76
Qwen2.5-Coder-7B w/SFT	54.9	48.8	54.88	64.60	51.27	**62.11**	68.55	60.76	33.54	**65.22**	57.62
Qwen2.5-Coder-7B w/DFT	**67.7**	**59.8**	**67.68**	**67.70**	**54.43**	60.87	**70.44**	**65.19**	**48.73**	63.35	**62.30**

Table 4: Performance comparison across different multi-modal reasoning benchmarks. The best performance on each benchmark is highlighted in bold.

表 4：不同多模态推理基准的性能比较。每个基准的最佳性能以粗体突出显示。

	MathVerse				MathVision	WeMath
	Vision Only	**Vision Intensive**	**Vision Dominant**	**Overall**		
Qwen2.5-VL-3B	28.81	30.96	31.60	33.83	21.25	4.10
Qwen2.5-VL-3B w/SFT	30.96	33.63	32.74	35.66	21.02	23.33
Qwen2.5-VL-3B w/DFT	**32.49**	**35.91**	**33.50**	**37.54**	**22.30**	**23.71**

and MultiPL-E (Cassano et al., 2023). Training is conducted for one epoch with a learning rate of 5×10^{-5}, a warm-up ratio of 0.05, and a batch size of 16.

并且使用 MultiPL-E（Cassano 等，2023）。训练使用一个 epoch，学习率为 5×10^{-5}，warm-up 比例为 0.05，批量大小为 16。

4.3.2 Results

4.3.2 结果

Table 3 shows DFT achieves improvements in most cases compared to both base models and SFT. For Qwen2.5-3B, DFT raises HumanEval from 43.3 to 45.7 and HumanEval+ from 36.0 to 39.0, with the MultiPL-E average also increasing from 40.05 (base) and 39.10 (SFT) to 41.84. Simi-lar trends are observed for Qwen2.5-Coder-3B, where DFT improves HumanEval to 56.7 and Hu-manEval+ to 50.0, outperforming both base and SFT. For Qwen2.5-Coder-7B, DFT reaches 67.7 on HumanEval, 59.8 on HumanEval+, and 62.3 average on MultiPL-E, surpassing SFT by +12.8, +11.0, and +4.7 points respectively. The overall trend demonstrates that DFT generally provides stronger performance across different models and languages.

表 3 显示，与基线模型和 SFT 相比，DFT 在大多数情况下都实现了性能提升。对于 Qwen2.5-3B，DFT 将 HumanEval 从 43.3 提升至 45.7，将 HumanEval+从 36.0 提升至 39.0，同时 MultiPL-E 的平均得分也从基线的 40.05 和 SFT 的 39.10 提升至 41.84。对于

Qwen2.5-Coder-3B，也观察到了类似的趋势，其中 DFT 将 HumanEval 提升至 56.7，将 HumanEval+提升至 50.0，优于基线和 SFT。对于 Qwen2.5-Coder-7B，DFT 在 HumanEval 上达到 67.7，在 HumanEval+上达到 59.8，在 MultiPL-E 上达到 62.3 的平均得分，分别比 SFT 高出 12.8、11.0 和 4.7 分。总体趋势表明，DFT 在不同模型和语言上通常能提供更强的性能。

4.4 Exploratory Experiment - Multi-Modal Reasoning

4.4 探索性实验 – 多模态推理

To further investigate the generalization of DFT, we extend our evaluation to multi-modal reasoning tasks that require both visual understanding and mathematical problem solving.

为进一步研究 DFT 的泛化能力，我们将评估范围扩展到需要同时具备视觉理解和数学问题解决能力的多模态推理任务。

4.4.1 Setup and Implementation Details

4.4.1 设置与实现细节

We use the WeThink dataset (Yang et al., 2025) for training. The model is fine-tuned using LLaMA-Factory (Zheng et al., 2024) and evaluated with VLMEvalKit (Duan et al., 2024). We train the model for 1 epoch with a learning rate of 5e-5. To comprehensively assess reasoning capabilities, we adopt a suite of multi-modal reasoning benchmarks including MathVerse (Zhang et al., 2024a), MathVision (Wang et al., 2024), and WeMath (Qiao et al., 2024) for evaluation.

我们使用 WeThink 数据集（Yang 等，2025）进行训练。模型使用 LLaMA-Factory（Zheng 等，2024）进行微调，并使用 VLMEvalKit（Duan 等，2024）进行评估。我们以学习率 5e-5 训练模型 1 个 epoch。为了全面评估推理能力，我们采用了一系列多模态推理基准测试，包括 MathVerse（Zhang 等，2024a）、MathVision（Wang 等，2024）和 WeMath（Qiao 等，2024）进行评估。

4.4.2 Results

4.4.2 结果

DFT achieves consistent improvements over base models and SFT across all multi-modal reasoning benchmarks. Table 4 shows, on MathVerse, DFT boosts Qwen2.5-VL-3B from 33.83 to 37.54 average accuracy, outperforming the SFT gain of only +1.83 by +3.71 points. Consistent improvements are observed across all major vision-related subcategories. On MathVision, DFT improves performance from 21.25 (base) to 22.30, exceeding SFT which fails to provide gains (21.02). On

DFT 在所有多模态推理基准测试中均比基线模型和 SFT 取得了持续改进。表 4 显示，在 MathVerse 上，DFT 将 Qwen2.5-VL-3B 的平均准确率从 33.83 提升至 37.54，优于 SFT 仅提升+1.83 的水平，高出+3.71 分。在所有主要的与视觉相关的子类别中均观察到持续的改进。在 MathVision 上，DFT 将性能从 21.25（基线）提升至 22.30，超过了 SFT 未能提供提升的 21.02。

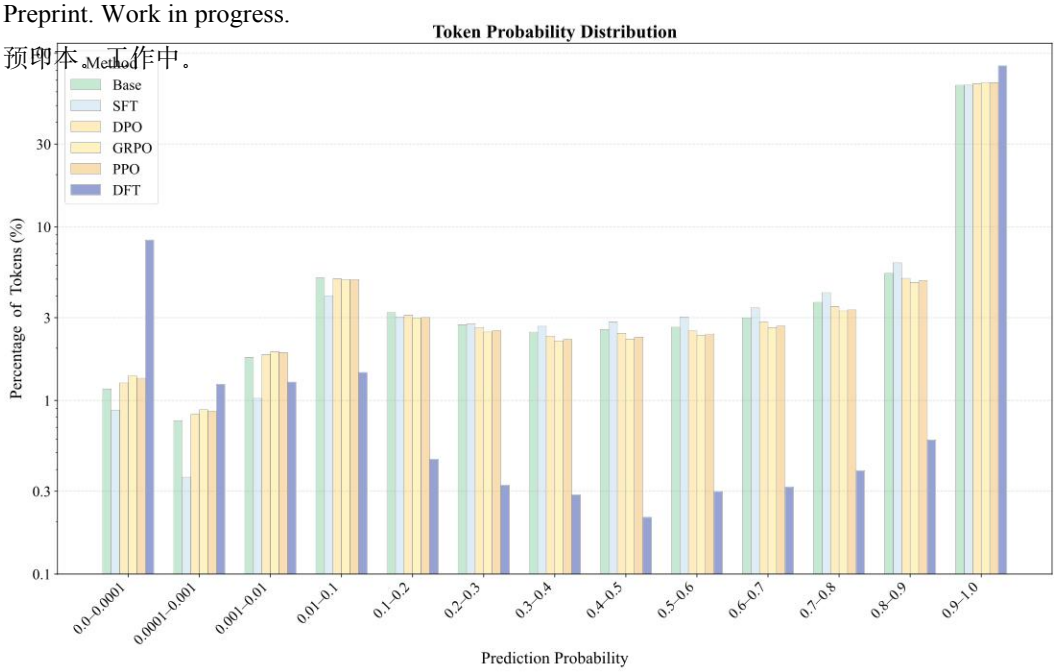

Figure 2: Token probability distributions on the training set before training and after fine-tuning with DFT, SFT, and various RL methods. A logarithmic scale is used on the y-axis for clarity.

图 2：在训练前和使用 DFT、SFT 以及各种强化学习方法微调后，训练集上的令牌概率分布。为了清晰起见，y 轴使用对数刻度。

WeMath, SFT already yields a +19.23 point gain, but DFT pushes performance slightly further to 23.71, maintaining superiority over both base and SFT. These results indicate that DFT not only strengthens text-only reasoning but also extends effectively to multi-modal domains.

WeMath，SFT 已经实现了 +19.23 分的提升，但 DFT 进一步提升了性能至 23.71 分，保持了相对于基线和 SFT 的优势。这些结果表明，DFT 不仅增强了纯文本推理能力，还有效地扩展到了多模态领域。

4.5 Analysis

4.5 分析

To understand how the model trained by DFT is different from SFT and other RL methods, we look into the token probability distribution of the model's output over the training set in Figure 2. SFT tends to uniformly increase token probabilities, shifting the entire distribution towards higher confidence, but mainly targeting the lower and lowest probability tokens. The highest probability token portion barely increases. In stark contrast, DFT exhibits a polarizing effect: it significantly boosts the probabilities of a subset of tokens while actively suppressing the probabilities of oth-ers. This leads to a bimodal distribution, with more tokens occupying both the highest and lowest probability bins. Other RL methods such as DPO, GPPO and PPO show the same trend as DFT, although the scale is much milder than it. We look into the words that belong to the lowest proba-

bility bin, and find that they are generally the conjunctive words or punctuations such as ‘the’, ‘let’,‘,’, ‘.’ etc. These results suggest that for robust learning, models should not attempt to fit all tokens with uniform confidence. It may be beneficial to deprioritize fitting tokens that serve grammatical functions rather than carrying primary semantic content. This concept is analogous to human peda-gogy, where students are taught to focus on substantive concepts rather than perfecting the usage of common connective words. Further analysis can be found in Appendix A.4.

为了理解通过 DFT 训练的模型与 SFT 及其他强化学习方法之间的差异，我们研究了图 2 中模型在训练集上输出的标记概率分布。SFT 倾向于均匀地增加标记概率，将整个分布向更高置信度的方向移动，但主要针对的是较低和最低概率的标记。最高概率部分的标记概率几乎没有增加。相比之下，DFT 表现出一种极化效应：它显著提升了一部分标记的概率，同时主动抑制其他标记的概率。这导致了双峰分布，更多的标记占据了最高和最低概率的区间。其他强化学习方法如 DPO、GPPO 和 PPO 也表现出与 DFT 相似的趋势，尽管其程度要轻得多。我们研究了属于最低概率区间的单词，发现它们通常是连接词或标点符号，如“the”、“let”、“,”、“.”等。这些结果表明，为了实现稳健的学习，模型不应试图以相同的置信度拟合所有标记。可能有益的是，优先考虑那些承载主要语义内容而非语法功能的标记。这一概念类似于人类教学，学生被教导关注实质性的概念，而不是完善常用连接词的使用。进一步的分析见附录 A.4。

5 CONCLUSION

5 结论

In this work, we address the well-documented generalization gap between Supervised Fine-Tuning and Reinforcement Learning. We provide a novel theoretical analysis, demonstrating that the standard SFT gradient is equivalent to a policy gradient update with an ill-posed implicit reward that is inversely proportional to the model’s confidence. This insight explains SFT’s tendency to overfit and its unstable optimization dynamics. Building on this analysis, we introduce DFT, a simple yet powerful method that rectifies this issue by dynamically reweighting the SFT loss with the token probability. This one-line modification stabilizes the learning process and promotes better generalization. Our extensive experiments show that DFT consistently outperforms standard SFT across

在本工作中，我们探讨了监督微调与强化学习之间众所周知的泛化差距。我们提供了一种新颖的理论分析，证明了标准的 SFT 梯度等价于一个隐式奖励不明确的策略梯度更新，该奖励与模型的置信度成反比。这一见解解释了 SFT 倾向于过拟合以及其优化动态不稳定的现象。基于这一分析，我们引入了 DFT，这是一种简单而强大的方法，通过动态地利用 token 概率重新加权 SFT 损失来解决这一问题。这一行修改稳定了学习过程，并促进了更好的泛化。我们的大量实验表明，DFT 在多个方面都优于标准的 SFT。

various models and challenging mathematical reasoning benchmarks. Furthermore, when adapted to an offline RL setting, DFT surpasses established online and offline RL algorithms, highlighting its effectiveness and efficiency. Our work provides both a deeper understanding of SFT and a practical, high-impact solution that significantly closes the performance gap with more complex RL methods.

各种模型和具有挑战性的数学推理基准。此外，当适应到离线强化学习（offline RL）环境中时，DFT 超越了现有的在线和离线强化学习算法，突显了其有效性和效率。我们的工作不仅加深了对 SFT 的理解，还提供了一个实用且具有高影响力的解决方案，显著缩小了与更复杂强化学习方法之间的性能差距。

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A APPENDIX

附录

A.1 USAGE OF LLM

A.1 LLM 的使用

We employ LLM primarily as writing assistants to refine and polish the manuscript. Their usage was limited to improving clarity, coherence, and presentation, while all conceptual and experimental contributions remain original.

我们主要将 LLM 用作写作助手，以润色和修改手稿。其使用仅限于提高清晰度、连贯性和呈现效果，而所有概念性和实验性贡献仍保持原创。

A.2 LIMITATIONS

A.2 局限性

While our experiments demonstrate the effectiveness of DFT on mathematical reasoning benchmarks and code generation tasks, the evaluation scope remains limited. We have not yet assessed its performance on broader task categories or with larger-scale LLM, which we leave for future exploration. Moreover, DFT may not offer universal benefits across all scenarios. In domains that primarily involve the acquisition of factual knowledge, conventional SFT may remain the most efficient approach. Our aim is not to assert that DFT universally outperforms SFT, but rather to offer a new perspective on objective design by analyzing the distinction between RL and SFT.

尽管我们的实验表明 DFT 在数学推理基准和代码生成任务中是有效的，但评估范围仍然有限。我们尚未评估其在更广泛任务类别或更大规模的 LLM 上的表现，这我们留待未来进一步探索。此外，DFT 可能并非在所有场景下都能提供普遍的优势。在主要涉及事实性知识获取的领域中，传统的 SFT 可能仍然是最有效的方法。我们的目标并非断言 DFT 在所有情况下都优于 SFT，而是通过分析强化学习与 SFT 之间的区别，提供一种新的客观设计视角。

A.3 DETAILED DERIVATION OF EQUATION (5)

A.3 方程（5）的详细推导

We start from the SFT gradient in Equation (2):

我们从方程(2)中的 SFT 梯度开始：

$$\nabla_\theta L_{\mathrm{SFT}}(\theta) = \mathrm{E}_{(x,y_\star)\sim D} - \nabla_\theta \log \pi_\theta(y_\star \mid x) . \qquad (1)$$

∇ θ LSFT(θ) = E_{(x,y⋆)~D} [−∇ θ log π θ (y⋆| x)] . (1)

For each query x, the expectation over expert demonstrations $(x, y_\star)$ can be written explicitly as a summation over all possible outputs y:

对于每个查询 x，专家示范（x, y⋆）上的期望可以显式地表示为所有可能输出 y 的求和：

$$\mathbb{E}_{(x,y_\star)\sim D} - \nabla_\theta \log \pi_\theta(y_\star \mid x) = \mathbb{E}_{x\sim D_x} \sum_y \mathbf{1}[y = y_\star] \Big] - \nabla_\theta \log \pi_\theta(y \mid x) . \quad (2)$$

E(x,y⋆)~D−∇θ log π θ (y⋆| x) = Ex~Dx [1[y = y⋆]]−∇θ log π θ (y | x) . (2)

We insert the model distribution $\pi_\theta(y \mid x)$, which allows us to express the summation in terms of importance weights:

我们插入模型分布 π θ (y | x)，这使我们能够将求和表示为重要性权重的形式：

$$\mathbb{E}_{x\sim D_x} \sum_y \pi_\theta(y \mid x) \cdot \frac{\mathbf{1}[y = y_\star]}{\pi_\theta(y \mid x)} \Big] - \nabla_\theta \log \pi_\theta(y \mid x) . \quad (3)$$

$ \mathbb{E}_{x \sim \mathcal{D}_x} \left[\pi_\theta(y^* | x) - \nabla_\theta \log \pi_\theta(y^* | x) \right] $. (3)

Here, the term $\mathbf{1}[y=y_\star]/\pi_\theta(y|x)$ serves as an importance weight comparing the expert (Dirac delta) distribu-tion with the model's distribution.

在这里，项 1[y=y⋆]/π θ (y|x)作为重要性权重，用于比较专家（狄拉克 δ）分布与模型分布。

The summation over y can now be rewritten as an expectation under the policy distribution $y \sim \pi_\theta(\cdot \mid x)$:

$$\mathbb{E}_{x\sim D_x} \mathbb{E}_{y\sim\pi_\theta(\cdot|x)} \frac{\mathbf{1}[y = y_\star]}{\pi_\theta(y \mid x)} \Big] - \nabla_\theta \log \pi_\theta(y \mid x) \quad (4)$$

现在可以将对 y 的求和重写为在策略分布 y ~ π θ (・ | x) 下的期望：

$$
\mathbb{E}_{x \sim D_x} \left[\mathbb{E}_{y \sim \pi_\theta(\cdot|x)} \left[\frac{1[y = y^\star]}{\pi_\theta(y|x)} \nabla_\theta \log \pi_\theta(y|x) \right] \right] \quad (4)
$$

Thus, we obtain Equation (5):

因此，我们得到方程（5）：

$$\mathbb{E}_{(x,y_\star)\sim D} - \nabla_\theta \log \pi_\theta(y_\star \mid x) = \mathbb{E}_{x\sim D_x} \mathbb{E}_{y\sim\pi_\theta(\cdot|x)} \frac{\mathbf{1}[y = y_\star]}{\pi_\theta(y \mid x)} \Big] - \nabla_\theta \log \pi_\theta(y \mid x) . \quad (5)$$

E(x,y⋆)~D−∇θ log π θ (y⋆| x) = Ex~Dx Ey~π θ (・|x) [1[y = y⋆]/π θ (y | x)] −∇θ log π θ (y | x) . (5)

This derivation shows that the SFT gradient can be expressed as an on-policy policy gradient with importance sampling, where the expert demonstration distribution is reweighted relative to the model distribution.

这个推导表明，SFT 梯度可以表示为带有重要性采样的在线策略梯度，其中专家演示分布相对于模型分布进行了重新加权。

A.4 Discussions and Insights

A.4 讨论与见解

Gradient Analysis of DFT. We now analyze the gradient induced by the DFT surrogate loss. Recall the sequence-level definition:

DFT 的梯度分析。我们现在分析 DFT 替代损失诱导的梯度。回想序列级别的定义：

$$L_{\mathrm{DFT}}(\theta) = -\mathrm{sg}\pi_\theta(y_\star|\, x) \log \pi_\theta(y_\star|\, x), \tag{10}$$

where sg(·) denotes the stop-gradient operator. Since the stop-gradient blocks backpropagation, the detached probability $\mathrm{sg}(\pi_\theta(y_\star|x))$ is treated as a constant during differentiation. Consequently, the gradient becomes

其中 sg(・) 表示 stop-gradient 操作符。由于 stop-gradient 会阻止反向传播，因此在求导过程中，分离的概率 sg(π θ (y⋆| x)) 被视为常数。因此，梯度变为

$$\nabla_\theta L_{\mathrm{DFT}} = -\mathrm{sg}\left(\pi_\theta(y_\star|x)\right)\frac{1}{\pi_\theta(y_\star|x)}\nabla_\theta \pi_\theta(y_\star|x) \tag{11}$$

$$= -\frac{\mathrm{sg}(\pi_\theta(y_\star|x))}{\pi_\theta(y_\star|x)}\nabla_\theta \pi_\theta(y_\star|x) \tag{12}$$

Since $\mathrm{sg}(\pi_\theta(y_\star|x))$ equals $\pi_\theta(y_\star|x)$ in the forward pass, the prefactor is numerically equal to 1.

由于在前向传播中 sg(π θ (y⋆| x))等于 π θ (y⋆| x)，因此前因子在数值上等于 1。

Therefore,

$$\nabla_\theta L_{\mathrm{DFT}} = -\nabla_\theta \pi_\theta(y_\star|x). \tag{13}$$

因此，

∇ θ LDFT = −∇ θ π θ (y⋆| x). (13)

This shows that DFT is mathematically equivalent to directly maximizing the model probability of the target token, rather than its log-probability as in cross-entropy.

这表明 DFT 在数学上等价于直接最大化目标标记的模型概率，而不是交叉熵中使用的对数概率。

For cross-entropy, the loss is

$$L_{\mathrm{CE}}(\theta) = -\log \pi_\theta(y_\star|x),$$

交叉熵的损失为

$ L_{CE}(\theta) = -\log \pi_\theta(y^\star| x) $

yielding gradient

$$\nabla_\theta L_{\mathrm{CE}} = -\frac{1}{\pi_\theta(y_\star|x)}\nabla_\theta \pi_\theta(y_\star|x).$$

梯度为

∇ θ LCE = − π θ (y⋆|x)∇ θ π θ (y⋆|x)。

Thus both CE and DFT share the same gradient direction but differ in scaling: CE amplifies updates for low-probability targets (factor $1/\pi$), while DFT applies a uniform factor 1. As a result, DFT avoids the instability caused by excessively large gradients on unlikely expert tokens, providing more conservative and stable updates.

因此，CE 和 DFT 具有相同的梯度方向，但在缩放方面有所不同：CE 对低概率目标的更新进行放大（因子为 1/π），而 DFT 则应用统一的因子 1。因此，DFT 避免了由于在不太可能的专家标记上出现过大梯度而引起的不稳定性，提供了更加保守和稳定的更新。

From the reinforcement learning perspective, the reward of DFT becomes uniformly 1 across all expert trajectories, equivalent to a verification-style objective that treats all correct references equally. From the optimization perspective, DFT trades off aggressive fitting of rare tokens for better stabil-

ity and calibration. Practically, this explains why DFT often yields smoother training and stronger generalization, while maintaining alignment with the pre-training distribution.

从强化学习的角度来看，DFT 的奖励在所有专家轨迹上变为统一的 1，等同于一种验证风格的目标，该目标对所有正确的参考进行平等对待。从优化的角度来看，DFT 在稀有标记的激进拟合与更好的稳定性和校准之间进行权衡。实际上，这解释了为什么 DFT 通常能产生更平滑的训练和更强的泛化能力，同时保持与预训练分布的一致性。

Learning from Noisy Data. DFT offers a simple yet effective approach, prompting us to reflect on why it might actually work. One intuitive explanation lies in its ability to learn from noisy data (Fre-und, 2009). Sasaki & Yamashina (2020) propose an imitation learning algorithm for learning from noisy demonstrations, based on the core idea of avoiding the fitting of data that is difficult to model, as such data is likely to originate from suboptimal behaviors, i.e., noise. Their method introduces a weighted behavioral cloning objective, where the weights are derived from a previously trained policy's confidence in each action. Similarly, the weighting mechanism in DFT shares the same intuition, but instead of relying on a fixed old policy model to compute confidence scores, it uses a single policy model to perform confidence-based weighting on-the-fly during training.

从噪声数据中学习。DFT 提供了一种简单而有效的途径，促使我们思考它为何可能有效。一个直观的解释在于其能够从噪声数据中学习（Freund, 2009）。Sasaki & Yamashina（2020）提出了一种模仿学习算法，用于从噪声演示中学习，其核心思想是避免拟合难以建模的数据，因为这类数据很可能来源于次优行为，即噪声。他们的方法引入了一个加权行为克隆目标，其中权重来源于先前训练策略对每个动作的信心。同样，DFT 中的加权机制也遵循相同的直觉，但与依赖于固定旧策略模型来计算置信度得分不同，它使用单一策略模型在训练过程中实时基于置信度进行加权。

A.5 Comparision with Concurrent work iw-SFT

A.5 与并发工作 IW-SFT 的比较

We include a concurrent method, Importance-Weighted SFT (iw-SFT) (Qin & Springenberg, 2025), for comparison. All training settings follow those reported in the original paper, except that we set the number of training epochs to 1.

我们引入了一个并行方法，重要性加权 SFT（iw-SFT）（Qin & Springenberg, 2025）进行比较。所有训练设置均遵循原始论文中报告的设置，只是我们将训练轮数设置为 1。

As shown in Table 5, DFT achieves higher average accuracy than iw-SFT on most model fami-lies: LLaMA-3.2-3B (+2.39), LLaMA-3.1-8B (+4.15), DeepSeekMath-7B (+3.34), and Qwen2.5-Math-1.5B (+1.30). Although iw-SFT outperforms our method on Qwen2.5-Math-7B (+2.45), this improvement is not consistent across datasets. In particular, for LLaMA-3.2-3B, iw-SFT under-performs standard SFT on Math500 (5.13 vs. 8.65) and AMC23 (2.03 vs. 3.13). Similarly, for LLaMA-3.1-8B, iw-SFT results in worse performance than SFT on Minerva Math (4.31 vs. 5.78) and AMC23 (7.34 vs. 8.28). In contrast, DFT consistently improves upon both the base model and SFT across nearly all datasets, including those where iw-SFT fails. These results underline better generalization ability of DFT in diverse mathematical reasoning scenarios. Moreover, iw-SFT incurs additional computational overhead by requiring a separate reference model to compute importance weights, whereas DFT dynamically derives its own weighting directly from the token probabilities of model, resulting in a more efficient training procedure.

如表 5 所示，DFT 在大多数模型家族中实现了比 iw-SFT 更高的平均准确率：LLaMA-3.2-3B（+2.39）、LLaMA-3.1-8B（+4.15）、DeepSeekMath-7B（+3.34）以及 Qwen2.5-Math-1.5B（+1.30）。尽管 iw-SFT 在 Qwen2.5-Math-7B 上优于我们的方法（+2.45），但这种改进在不同数据集上并不一致。特别是对于 LLaMA-3.2-3B，在 Math500 上 iw-SFT 的表现不如标准 SFT（5.13 vs. 8.65），在 AMC23 上也表现较差（2.03 vs. 3.13）。同样地，在 LLaMA-3.1-8B

上，iw-SFT 在 Minerva Math（4.31 vs. 5.78）和 AMC23（7.34 vs. 8.28）上的表现也比 SFT 差。相比之下，DFT 在几乎所有数据集上都一致地提升了基础模型和 SFT 的表现，包括那些 iw-SFT 无法处理的数据集。这些结果表明，DFT 在多样化的数学推理场景中具有更好的泛化能力。此外，iw-SFT 需要单独的参考模型来计算重要性权重，从而带来了额外的计算开销，而 DFT 则直接从模型的 token 概率动态推导自身的权重，从而实现更高效的训练过程。

Table 5: Comparison with concurrent work iw-SFT on math benchmarks. DFT outperforms the iw-SFT in most settings across model families and benchmarks. The best performance is bold.

表 5：与同时期工作 iw-SFT 在数学基准测试中的比较。DFT 在大多数模型家族和基准测试中均优于 iw-SFT。最佳性能以粗体显示。

	Math500	Minerva Math	Olympiad Bench	AIME24	AMC23	Avg.
LLaMA-3.2-3B w/iw-SFT	5.13	2.63	1.51	0.00	2.03	2.26
LLaMA-3.2-3B w/DFT	**12.79**	**2.84**	**2.90**	**0.83**	**3.91**	**4.65**
LLaMA-3.1-8B w/iw-SFT	18.21	4.31	4.31	0.20	7.34	6.87
LLaMA-3.1-8B w/DFT	**27.44**	**8.26**	**6.94**	**0.41**	**12.03**	**11.02**
DeepSeekMath-7B w/iw-SFT	35.32	8.75	11.11	0.61	**18.28**	14.81
DeepSeekMath-7B w/DFT	**41.46**	**16.79**	**15.00**	**1.24**	16.25	**18.15**
Qwen2.5-Math-1.5B w/iw-SFT	59.38	17.08	26.82	**8.13**	**40.00**	30.28
Qwen2.5-Math-1.5B w/DFT	**64.89**	**20.94**	**27.08**	6.87	38.13	**31.58**
Qwen2.5-Math-7B w/iw-SFT	**70.28**	25.70	**34.46**	**16.46**	**51.09**	**39.60**
Qwen2.5-Math-7B w/DFT	68.20	**30.16**	33.83	8.56	45.00	37.15

Table 6: Evaluation results on five mathematical reasoning benchmarks in the offline reinforcement learning setting using rejection-sampling–based reward signals, compared against iw-SFT.

表 6：在离线强化学习设置下，使用基于拒绝抽样的奖励信号对五个数学推理基准进行评估的结果，与 iw-SFT 进行比较。

	Setting	Math500	Minerva Math	Olympiad Bench	AIME24	AMC23	Avg.
Qwen2.5-Math-1.5B w/iw-SFT	SFT	59.38	17.08	26.82	8.13	40.00	30.28
Qwen2.5-Math-1.5B w/DFT	SFT	62.50	22.94	26.87	7.31	33.75	30.67
Qwen2.5-Math-1.5B w/iw-SFT	Offline	60.80	18.13	27.83	**8.33**	44.21	31.86
Qwen2.5-Math-1.5B w/DFT	Offline	**64.71**	**25.16**	**30.93**	7.93	**48.44**	**35.43**

We also compare against iw-SFT under the offline setting, as shown in Table 6. While iw-SFT performs competitively on certain datasets, achieving 60.80 on Math500 and 44.21 on AMC23, its overall average performance (31.86) remains below that of our method by +3.57 points. Moreover, iw-SFT shows only modest improvements compared to its standard SFT counterpart, with an average score of 31.86 in the offline RL setting versus 30.28 with SFT (+1.58). In contrast, DFT achieves a larger gain of +4.76 (from 30.67 to 35.43). These results indicate that iw-SFT provides limited benefits from reward supervision under offline constraints, whereas DFT is able to more effectively incorporate such signals, leading to better generalization and higher task performance.

我们还在离线设置下将我们的方法与 iw-SFT 进行了比较，如表 6 所示。虽然 iw-SFT 在某些数据集上表现相当不错，在 Math500 上达到 60.80，在 AMC23 上达到 44.21，但其整体平均性能（31.86）仍比我们的方法低+3.57 分。此外，与标准 SFT 相比，iw-SFT 的改进幅度有限，在离线强化学习设置下的平均得分是 31.86，而 SFT 的平均得分是 30.28（+1.58）。相比

之下，DFT 的提升更大，达到了+4.76（从 30.67 提升到 35.43）。这些结果表明，在离线约束下，iw-SFT 从奖励监督中获得的益处有限，而 DFT 则能够更有效地融合这些信号，从而实现更好的泛化能力和更高的任务表现。

A.6 Exploratory Experiment - OpenR1-Math Training Dataset

A.6 探索性实验 - OPENR1-MATH 训练数据集

Inspired by DeepSeek-R1 DeepSeek-AI et al. (2025), several studies have attempted to train open-source models to reproduce its reasoning capabilities (Hugging Face, 2025). To this end, a high-quality dataset, OpenR1-Math-220k (Hugging Face, 2025), was constructed, where the prompts are drawn from NuminaMath 1.5 and the off-policy reasoning traces are generated by DeepSeek-R1. LUFFY (Yan et al., 2025) further filtered out sequences longer than 8192 tokens as well as those verified incorrect by Math-Verify, resulting in about 45k prompts paired with off-policy reasoning traces. We adopt this dataset as the training corpus for SFT. All training details remain the same as previous experiments, except that the number of epochs is set to 3.

受 DeepSeek-R1 和 DeepSeek-AI 等（2025）的启发，一些研究尝试训练开源模型以复现其推理能力（Hugging Face，2025）。为此，构建了一个高质量的数据集 OpenR1-Math-220k（Hugging Face，2025），其中提示来源于 NuminaMath 1.5，离策略推理轨迹由 DeepSeek-R1 生成。LUFFY（Yan 等，2025）进一步过滤掉长度超过 8192 个 token 的序列以及 Math-Verify 验证为错误的序列，最终得到约 45k 个提示与离策略推理轨迹的配对。我们将此数据集作为 SFT 的训练语料库。所有训练细节与之前实验保持一致，唯一不同的是将训练轮数设置为 3。

As shown in Table 7, training on OpenR1-Math-220k consistently improves performance, and the use of higher-quality annotations yields additional gains. SFT on this dataset increases the average accuracy of Qwen2.5-Math-1.5B by +13.24 points compared to the base model, while DFT provides a further +9.03 gain, resulting in a total improvement of +22.27. These results suggest that DFT remains effective even when applied on top of high-quality training data, highlighting its potential as a general fine-tuning paradigm.

如表 7 所示，使用 OpenR1-Math-220k 进行训练可以持续提升性能，而使用高质量的标注数据还能带来额外的提升。在该数据集上进行监督微调（SFT），使 Qwen2.5-Math-1.5B 的平均准确率比基础模型提高了 13.24 个百分点，而数据集微调（DFT）在此基础上进一步提升了 9.03 个百分点，总提升达到了 22.27 个百分点。这些结果表明，即使在高质量训练数据上应用 DFT，其效果仍然显著，凸显了其作为通用微调范式的潜力。

Table 7: Performance training on the OpenR1-Math-220k dataset. The best score for each benchmark is in bold.

表 7：在 OpenR1-Math-220k 数据集上的性能训练。每个基准测试的最佳得分加粗显示。

	Math500	Minerva Math	Olympiad Bench	AIME24	AMC23	Avg.
Qwen2.5-Math-1.5B	31.66	8.51	15.88	4.16	19.38	15.92
Qwen2.5-Math-1.5B w/SFT	61.60	20.29	24.27	4.16	35.47	29.16
Qwen2.5-Math-1.5B w/DFT	**71.76**	**27.00**	**33.48**	**9.79**	**48.91**	**38.19**

Table 8: Performance on five mathematical reasoning benchmarks using LoRA for training. The best score of each model across benchmarks is highlighted in bold.

表 8：使用 LoRA 进行训练的五项数学推理基准测试的性能。每个模型在基准测试中的最佳得分用粗体突出显示。

	Math500	Minerva Math	Olympiad Bench	AIME24	AMC23	Avg.
LLaMA-3.2-3B	1.63	1.36	1.01	**0.41**	1.56	1.19
LLaMA-3.2-3B w/SFT	4.88	1.56	1.68	0.00	2.66	2.56
LLaMA-3.2-3B w/DFT	**11.13**	**5.18**	**3.87**	0.00	**2.97**	**4.63**
Qwen2.5-Math-1.5B	31.66	8.51	15.88	4.16	19.38	15.92
Qwen2.5-Math-1.5B w/SFT	41.47	10.85	11.56	1.45	17.03	16.87
Qwen2.5-Math-1.5B w/DFT	**64.85**	**22.58**	**28.45**	**5.84**	**40.78**	**32.90**

A.7 Exploratory Experiment - PEFT Training Setting

To investigate whether DFT remains effective under parameter-efficient fine-tuning (PEFT) settings with limited compute, we apply DFT using LoRA adapters across two model families: LLaMA-3.2-3B and Qwen2.5-Math-1.5B. All training configurations remain identical to previous full-parameter experiments, except that LoRA is enabled with rank=8 and alpha=16.

为了研究在计算资源有限的参数高效微调（PEFT）设置下，DFT 是否仍然有效，我们在两个模型家族 LLaMA-3.2-3B 和 Qwen2.5-Math-1.5B 上应用了使用 LoRA 适配器的 DFT。所有训练配置与之前的全参数实验保持一致，只是启用了 LoRA，且秩为 8，alpha 为 16。

As shown in Table 8, DFT provides consistent improvements over both base and SFT baselines under LoRA-based PEFT. For Qwen2.5-Math-1.5B, DFT increases the average accuracy from 15.92 (base) and 16.87 (SFT) to 32.90. For LLaMA-3.2-3B, DFT achieves a gain of +3.44 over SFT (from 1.19 to 4.63). These results indicate that DFT can serve as an effective fine-tuning strategy in low-resource or compute-constrained settings, where full model updates are not practical.

如表 8 所示，在基于 LoRA 的 PEFT 方法下，DFT 在基线模型和 SFT 基线模型上都提供了稳定的提升。对于 Qwen2.5-Math-1.5B 模型，DFT 将平均准确率从 15.92（基线）和 16.87（SFT）提升到了 32.90。对于 LLaMA-3.2-3B 模型，DFT 在 SFT 基础上提升了+3.44（从 1.19 提升至 4.63）。这些结果表明，DFT 可以作为一种有效的微调策略，在资源有限或计算受限的场景下，此时进行完整的模型更新并不实际。

A.8 Training Hyper-Parameters Ablation

A.8 训练超参数消融实验

To assess the robustness and sensitivity of our approach (DFT) with respect to key training hyper-parameters, we conduct an ablation study focused on learning rate and batch size, using the Qwen2.5-Math-1.5B base model. This analysis aims to answer two central questions: (1) Is the performance gap between DFT and SFT due to a suboptimal hyperparameter configuration in SFT? (2) How sensitive are both methods to changes in learning rate and batch size?

为了评估我们的方法（DFT）相对于关键训练超参数的鲁棒性和敏感性，我们使用 Qwen2.5-Math-1.5B 基础模型开展了一项消融研究，重点研究学习率和批量大小。该分析旨在回答两个核心问题：(1) DFT 与 SFT 之间的性能差距是否是由于 SFT 的超参数配置不佳所致？(2) 两种方法对学习率和批量大小变化的敏感性如何？

We evaluate both DFT and SFT across four learning rates: 2e-4, 1e-4, 5e-5, and 1e-5. As shown in Figure 3 (left), both methods exhibit a certain degree of sensitivity to the learning rate. DFT consistently outperforms SFT under all configurations, suggesting that the performance gap cannot be attributed solely to suboptimal hyperparameter choices in SFT. For both methods, intermediate learning rates (1e-4 and 5e-5) yield the best results, while both lower (1e-5) and higher (2e-4) values lead to noticeable degradation.

我们在四种学习率（2e-4、1e-4、5e-5 和 1e-5）下评估了 DFT 和 SFT。如图 3（左图）所示，两种方法在一定程度上对学习率敏感。DFT 在所有配置下都优于 SFT，这表明 SFT 的性能差距不能仅仅归因于其超参数选择不当。对于这两种方法，中间的学习率（1e-4 和 5e-5）取得了最佳效果，而较低的学习率（1e-5）和较高的学习率（2e-4）则导致了明显的性能下降。

We further assess the impact of batch size, sweeping values from 32 to 256. As shown in Figure 3 (right), both DFT and SFT exhibit relatively stable performance across the full range of batch sizes. While minor fluctuations are observed, there is no consistent trend indicating that larger or smaller batches significantly affect final accuracy. This suggests that batch size is not a dominant factor for either method in this setup, and that default values may suffice in practice.

我们进一步评估了批量大小的影响，将批量大小从 32 扫到 256。如图 3（右图）所示，在整个批量大小范围内，DFT 和 SFT 都表现出相对稳定的性能。虽然观察到一些微小的波动，但没有一致的趋势表明更大的或更小的批量显著影响最终的准确性。这表明，在这种设置下，批量大小对两种方法都不是一个主要因素，实际应用中默认值可能就足够了。

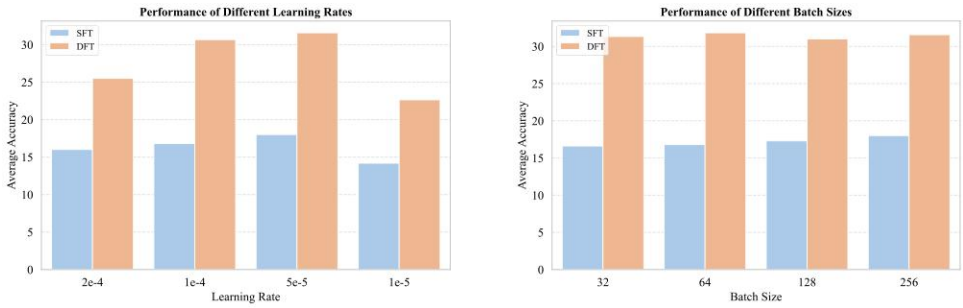

Figure 3: Ablation study of training hyper-parameters, learning rates and batch size, for DFT and SFT on Qwen2.5-Math-1.5B model.

图 3：在 Qwen2.5-Math-1.5B 模型上对 DFT 和 SFT 的训练超参数、学习率和批量大小进行消融研究。